

CSA1713 – Artificial Intelligence

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Slot – C

6.DEPTH FIRST SEARCH (DFS)

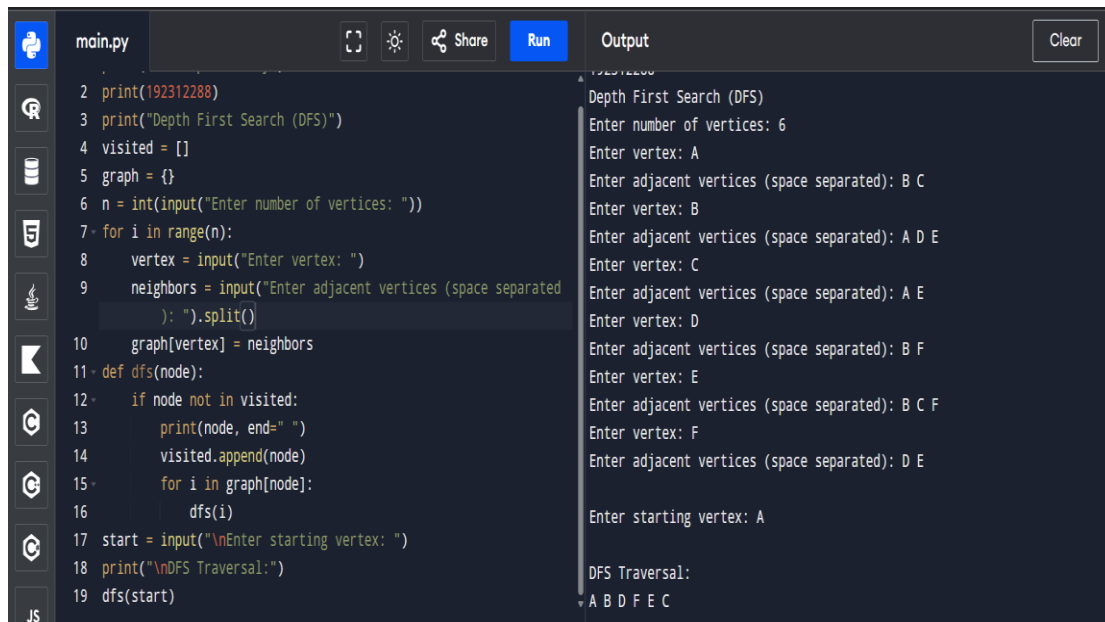
Code:

```
print("B. Deepthi Reddy")
print(192312288)
print("Depth First Search (DFS)")
visited = []
graph = {}
n = int(input("Enter number of vertices: "))
for i in range(n):
    vertex = input("Enter vertex: ")
    neighbors = input("Enter adjacent vertices (space separated): ").split()
    graph[vertex] = neighbors
def dfs(node):
    if node not in visited:
        print(node, end=" ")
        visited.append(node)
        for i in graph[node]:
            dfs(i)
start = input("\nEnter starting vertex: ")
```

```
print("\nDFS Traversal:")
```

```
dfs(start)
```

Output:



The screenshot shows a code editor with a Python file named `main.py`. The code implements a Depth First Search (DFS) algorithm on a graph. The output window shows the execution of the program, including user input and the resulting DFS traversal.

```
main.py
2 print(192312288)
3 print("Depth First Search (DFS)")
4 visited = []
5 graph = {}
6 n = int(input("Enter number of vertices: "))
7 for i in range(n):
8     vertex = input("Enter vertex: ")
9     neighbors = input("Enter adjacent vertices (space separated
    ): ").split()
10    graph[vertex] = neighbors
11 def dfs(node):
12     if node not in visited:
13         print(node, end=" ")
14         visited.append(node)
15         for i in graph[node]:
16             dfs(i)
17 start = input("\nEnter starting vertex: ")
18 print("\nDFS Traversal:")
19 dfs(start)
```

Output

```
192312288
Depth First Search (DFS)
Enter number of vertices: 6
Enter vertex: A
Enter adjacent vertices (space separated): B C
Enter vertex: B
Enter adjacent vertices (space separated): A D E
Enter vertex: C
Enter adjacent vertices (space separated): A E
Enter vertex: D
Enter adjacent vertices (space separated): B F
Enter vertex: E
Enter adjacent vertices (space separated): B C F
Enter vertex: F
Enter adjacent vertices (space separated): D E

Enter starting vertex: A

DFS Traversal:
A B D F E C
```